9 Jean invests £12 000 in an account paying compound interest for 2 years.

In the first year the rate of interest is $x\%$
At the end of the first year the value of Jean's investment is £12 336

In the second year the rate of interest is $\frac{x}{2}\%$

What is the value of Jean's investment at the end of 2 years?

First-year interest = 12336 − 12000 = £336.

GBP = pounds; Pct = percentage of the starting £12000.

	GBP	Pct	
	12000	100	
÷ 100			÷ 100
	120	1	
× 2.8			× 2.8
	336	2.8	

1 mark
Compare the interest with the original amount

1 mark
First-year rate x = 2.8%

Second-year rate = 2.8 ÷ 2 = 1.4%.

Use the new balance: multiplier = 1 + 1.4 ÷ 100 = 1.014.

12336 × 1.014 = 12508.704

1 mark
Apply the second-year multiplier to £12336

1 mark
Round the final balance to the nearest penny

£ *12508.70*

(Total for Question 9 is 4 marks)